Engineering Mathematics - I · Unit_04 · Unit 4

DRAFT — UE26MA141A · Unit_04 · Unit 4 · Problems on Beta an Gamma functions

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Problems on Beta an Gamma functions
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Problems on Beta an Gamma functions

1 Problems on Beta an Gamma functions

Study focus: Problems on Beta an Gamma functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In previous few classes, you have studied definition and properties of beta gamma functions, relationship between beta gamma function and duplication formula.

Key details

- In today's class, we are going to solve some problems on beta gamma function.
- Before that, we will recall some of the formulas you have studied earlier, which are useful in solving problems for today's class.
- First one is gamma n is equal to integral 0 to infinity e to the power minus x power n minus 1 dx.
- This is definition of gamma function.
- Second one is gamma of n plus 1 is equal to n gamma n.
- Third one is gamma half is equal to square root of pi.
- Porth one is beta of m comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 1 dx.
- Next one is integral zero to pi by two sine power p theta cos power q theta t theta is equal to one by two times beta of p plus one by two comma q plus one by two, which is derived from the previous result itself.
- And the last one, but very important one, that is beta of m, comma n is equal to gamma m into gamma n divided by gamma of m plus n.
- That is the relationship between beta gamma function.
- Lear these formulas, all these formulas are very important.
- If minus log x is equal to t, x will be e to the power minus t, and dx is minus a power minus t dt.
- Therefore, when x is equal to 0, we have t is equal to log of 1 by 0.

- Thus you have when x is 0, t is equal to infinity, and when x is equal to 1, we have t is equal to log of 1 divided by 1, that is log 1 and log 1 is equal to 0.
- Therefore, integral 0 to 1 dx divided by square root of minus log x is equal to integral infinity to 0 minus a power minus d dt divided by square root of t.
- which is equal to integral 0 to infinity e to the power minus t.
- And minus of integral a to b is integral b to a so it is integral zero to infinity a power minus t to the power minus half dt.
- Integral zero to infinity e to the power minus t to the power minus one by two d we can write it as integral zero to infinity e power minus t to the power one by two minus one dt.
- And we have already studied the result that gamma half is equal to square root of pi.
- True that m plus 1 whole to the power n plus 1 into minus 1 power n into integral 0 to 1 x power m log x to the power n dx is equal to gamma of n plus 1.

Example or application from the lecture

- You can also consider the result gamma of n plus one is equal to n factorial when n is a positive integer which is derived from this result.
- Consider the first problem evaluate integral 0 to 1 dx divided by square root of minus log x.
- Which is log infinity, which you can consider it as infinite.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

where n is a positive integer and m is greater than minus.

Key details

- In this integral, let me make some substitution so that I will get gamma function.
- In gamma function, we need an exponential term.
- When x is equal to e to the power minus y, dx is equal to minus of e to the power minus y, dy.
- log x is equal to minus y or we can say minus of log x is equal to y or we have log of 1 divided by x is equal to y which is.
- Similar to the previous problem, where also we got the same limits of integration.
- So here also when x is equal to 0 that means y is equal to log of 1 by 0 or log infinity which is infinity.
- When x is equal to 1, we have y is equal to log of 1 by 1 and log of 1 by 1 is log 1 and log one is equal to zero.
- log of e to the power minus y is.
- Minus y so it is minus y to the power n and in the last term we have already taken the negative sign outside so it is e to the power minus y d y.
- And now minus of integral infinity to 0, we can write it as integral 0 to infinity.
- And from this term, minus y whole to the power n, I have taken.
- Minus 1 to the power n outside.
- How we got this is.

- From the previous step, here you have e to the power minus m y, here you have e to the power minus y.
- Therefore, integral 0 to 1 x power m log x all power n dx is equal to minus 1 power n integral 0 to infinity e to the power minus u because we have taken m plus 1 y is equal to u into.
- Y power n y is u by m plus one.
- Evaluate integral 0 to 1 x to the power 4 1 minus x whole q dx.

Example or application from the lecture

- To prove this result, first let me consider the integral 0 to 1 x power m log x to the power n dx.
- Then m plus 1 into dy is equal to du and also you can consider here y is equal to u divided by m plus 1.
- Consider the left hand side that is integral 0 to 1 x power 4 1 minus x whole cube dx that can be written as x to the power 5 minus 1 into 1 minus x whole to the power 4 minus 1 dx.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

Like this, just to compare this with the beta function definition, which is beta of m, comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 1 dx.

Key details

- And now you can use the relationship between beta and gamma function which is beta of m comma n.
- If you want to write it in general, you can write it for beta of.
- M comma n which is equal to gamma of m into gamma of n divided by gamma of m plus n.
- Gamma four is three factorial and gamma nine is eight factorial.
- Where you have 1 minus x power n whole to the power r.
- But we want to write the integral on the left hand side in terms of beta function.
- And beta function definition says that beta of m comma n is equal to integral 0 to 1 x to the power m minus 1 into 1 minus x to the power n minus 20x.
- But in this problem we have x to the power n here.
- So to prove this result, first take x power n is equal to y.
- Then n into x power n minus 1 dx is equal to dy.
- With this substitution, the limits of integration will not change because when x is equal to 0, y is 0.
- When x is equal to 1, y is also equal to 1.
- What is dx? dx is equal to.
- From the substitution it is dy divided by n into x power n minus 1, which is nothing but n into y power n minus 1 divided by n.
- That is equal to 1 by n, we can keep it outside the integral sign, integral 0 to 1.
- and into 1 divided by y to the power n minus 1 divided by n.

- which is equal to y to the power m by n divided by y to the power 1 minus 1 by n, which is equal to y to the power m by n minus 1 plus 1 by n.
- So because now this integral, integral 0 to 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

Y to the power m minus n plus one divided by n you can write it as m plus one divided by n minus one into one minus phi whole to the power r.

Key details

- R you can write it as r plus one minus one.
- Beta of m comma n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Beta an Gamma functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Beta an Gamma functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta an Gamma functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/004_Problems_on_Beta_an_Gamma_functions__ccd478a3bed2d97a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice